

Models for Multivariate Ordinal Data

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In previous modules, we explored what ordinal data are and how they can be modeled using approaches such as Proportional Odds Models and CUB models. These methods are well-suited for analyzing one or a few ordinal variables, especially when the interest lies in explaining or predicting an outcome. However, in many real-world settings, researchers are faced with multivariate ordinal data; that is, multiple ordinal variables collected simultaneously on each individual or unit.

Think, for example, of a customer satisfaction survey where participants rate a range of services or products. Or consider a psychological questionnaire measuring various aspects of well-being or attitude. In such cases, the goal is often not to model a single response, but rather to uncover patterns, similarities, or latent structures in the data. This calls for multivariate methods.

Before we delve into specific multivariate methods, it's helpful to briefly define two fundamental goals in multivariate data analysis that we will address:

- **Dimensionality Reduction:** Imagine you have a dataset with a very large number of variables, or “dimensions”, for each observation. This can make the data difficult to understand, visualize, and analyze. Dimensionality reduction is the process of transforming high-dimensional data into a lower-dimensional space (fewer variables) while trying to retain as much of the original, important information and relationships as possible. The

aim is to simplify the data, making it more manageable for visualization, interpretation, and subsequent analyses, often by uncovering underlying “latent” variables or themes.

- **Clustering:** In contrast to dimensionality reduction, clustering is an unsupervised learning technique aimed at finding natural groupings or segments within a dataset. It involves organizing a set of observations (e.g., individuals, items) into groups (called “clusters”) such that observations within the same group are more similar to each other than to those in other groups. Clustering helps to uncover hidden patterns, identify distinct segments, and simplify complex datasets by categorizing observations based on their inherent similarities, without any prior knowledge of what those groups might be.

These two concepts, though distinct, often complement each other in data analysis workflows. Dimensionality reduction can make data more amenable to clustering, for instance, by projecting it into a space where true similarities are more apparent.

While classical multivariate techniques, such as Principal Component Analysis (PCA) and K-means clustering, are powerful tools for continuous data, they encounter significant limitations when applied to ordinal data. The core challenge lies in the inherent nature of ordinal scales: their numeric labels represent ordered categories rather than true measurable quantities with uniform intervals. Respecting this fundamental distinction is paramount to avoid drawing misleading or erroneous conclusions from our analyses. In this chapter, we will explore several alternative approaches that are better suited to analyzing both multivariate ordinal data and mixed-type datasets (those combining nominal, ordinal, and continuous variables).

3.1 Why Classical Multivariate Methods Fall Short

The inadequacy of classical multivariate methods for ordinal data stems from their underlying assumptions, which are typically violated by the properties of ordinal scales.

3.1.1 Principal Component Analysis (PCA)

Classical PCA is a dimensionality reduction technique built upon the covariance or correlation matrix of the variables. Its objective is to identify a set of orthogonal linear combinations of the original variables (principal components) that capture the maximum variance in the data. The method inherently assumes that the data are numerical and that the distances between values are meaningful and consistent.

When PCA is applied directly to ordinal data (even after assigning numerical labels like 1, 2, 3, etc.), it treats these labels as if they represent continuous, interval-scaled quantities. This creates a critical misrepresentation. For example, PCA would treat the numerical “distance” between “agree” (e.g., coded as 4) and “strongly agree” (e.g., coded as 5) as identical to the “distance” between “neutral” (e.g., coded as 3) and “agree” (e.g., coded as 4). As previously

discussed, these perceived “distances” on an ordinal scale are rarely equal in their true conceptual or psychological magnitude. By treating these non-uniform intervals as equal, PCA can generate principal components that distort the true underlying structure of the ordinal relationships, leading to potentially misleading interpretations of the latent dimensions.

3.1.2 K-Means Clustering

Similarly, standard clustering algorithms like K-means rely heavily on Euclidean distances to quantify similarity or dissimilarity between data points. Euclidean distance, by its very definition, assumes interval-scaled variables where differences between values are direct and comparable.

Applying K-means directly to numerically coded ordinal data leads to similar problems as with PCA. The calculated Euclidean distances between response profiles will be based on the arbitrary numerical assignments, not on the true, often unequal, conceptual distances between ordered categories. This can result in the formation of artificial clusters that do not genuinely reflect meaningful patterns or homogeneous groups within the ordinal data. Consequently, relevant patterns might be masked, or spurious groupings might emerge, hindering accurate segmentation and interpretation.

Therefore, when confronted with multivariate ordinal data, it becomes important to use analytical tools that account for their ordered nature.

3.2 Techniques for Dimensionality Reduction and Visualization of Ordinal Data

The methods discussed in this section—Nonlinear PCA, Correspondence Analysis, and Multiple Correspondence Analysis—all share a common overarching goal: to reduce the dimensionality of complex multivariate data and to provide insightful graphical representations of the relationships within the data. While they achieve this through different mechanisms and for slightly different data structures, they collectively aim to uncover latent structures and visualize patterns in a lower-dimensional space.

3.2.1 Nonlinear Principal Component Analysis (NLPCA)

Nonlinear Principal Component Analysis (NLPCA) represents a significant advancement over classical PCA, extending its framework to accommodate diverse data types, including ordinal variables. When applied to ordinal data, NLPCA addresses the core challenge by transforming the original categorical responses into optimal scores.

The fundamental idea behind NLPCA is to “translate” ordinal data into a numerical scale in a way that maximizes the variance explained by the resulting principal components, much like traditional PCA. However, unlike classical PCA, this “translation” is not fixed or arbitrary.

Instead, NLPKA iteratively determines the most appropriate numerical values (optimal scores) for each category. These optimal scores are derived such that they best preserve the order of the categories and maximize the explained variance in the transformed data, effectively capturing potentially nonlinear relationships.

By employing optimal scaling, NLPKA can effectively handle data that are a mix of continuous, nominal, and ordinal variables, offering a flexible and powerful dimensionality reduction technique for heterogeneous datasets. The output of NLPKA is a set of principal components that reflect the major sources of variability, but crucially, they do so while respecting the intrinsic nature of the ordinal data. NLPKA is particularly valuable when one suspects that the latent constructs driving the data exhibit nonlinear relationships or are multidimensional in a non-Euclidean sense.

3.2.2 Correspondence Analysis (CA)

Correspondence Analysis (CA) is a statistical technique primarily designed for exploring the association between two categorical variables, typically presented in a contingency table (a cross-tabulation). CA’s strength lies in its ability to graphically represent the relationships between the row and column categories in a common low-dimensional space, usually two dimensions, where their proximity visually reflects their association.

For example, if we have a contingency table cross-tabulating “Education Level” (an ordinal variable) with “Preferred News Source” (a nominal or ordinal variable), CA can visually reveal if specific education levels are strongly associated with particular news sources. Points representing categories that are close together on the CA map indicate a strong tendency to co-occur.

While CA is most commonly associated with nominal data, it can also be applied to ordinal variables. However, it’s important to note that in its standard form, CA does not explicitly leverage or enforce the order information of ordinal variables. Consequently, while CA can be highly useful for detecting general patterns and visualizing broad relationships in the data, it should be applied with caution when the precise ordinal nature and monotonicity of the data are central to the research question. Nevertheless, CA remains a valuable exploratory tool, offering useful insights into co-occurrence structures and showing which categories of different variables are more likely to appear together, particularly in the initial phases of data exploration.

3.2.3 Multiple Correspondence Analysis (MCA)

Multiple Correspondence Analysis (MCA) extends the principles of CA to analyze the relationships among more than two categorical variables simultaneously. This makes MCA particularly well-suited for large survey datasets where respondents provide answers across numerous items, often using Likert-type scales.

Similar to CA, MCA represents both the categories of the variables and the individual respondents in a low-dimensional space (e.g., a 2D scatter plot). This visual representation greatly facilitates the detection of latent structures and the visualization of similarities among individuals and variables. For instance, MCA can help identify clusters of individuals who exhibit similar response profiles across multiple survey questions. It can also reveal the underlying dimensions or “themes” that explain the relationships among the survey items themselves.

A key point to remember about standard MCA is that it traditionally treats all variables as nominal, thereby disregarding the inherent ordered nature of ordinal variables. To mitigate this limitation, several extensions of MCA have been proposed. These include methods like ordinal MCA or nonlinear MCA, which incorporate specific coding and weighting schemes for responses that explicitly account for their ordering. These advanced versions strive to preserve the monotonicity of ordinal scales and to produce more meaningful and interpretable dimensions that are faithful to the data’s original structure.

In practice, MCA is widely employed in the social sciences, marketing research, and psychology due to its powerful ability to simplify complex datasets into interpretable patterns. It is an invaluable tool for dimensionality reduction and visualization, often preceding or complementing other analyses by providing a simplified, meaningful representation of the data.

3.3 Clustering with Ordinal and Mixed Data

Clustering is a milestone of unsupervised learning since it allows to discover natural groupings or segments within a dataset. However, as anticipated, standard clustering algorithms like K-means are poorly suited for ordinal or mixed-type data due to their reliance on distance metrics that assume continuous, interval-scaled variables.

3.3.1 Distance-Based Clustering

To perform clustering on ordinal data, the critical first step is to define distance measures that genuinely respect the ordinal nature of the variables. One of the most widely adopted and versatile approaches is to use Gower’s distance (or Gower’s coefficient of similarity). Gower’s distance can handle mixed data types, including nominal, ordinal, and continuous variables, within a single dissimilarity metric. For ordinal variables specifically, Gower’s metric calculates distances in a way that preserves the order and accounts for the number of categories separating two responses, without assuming equal intervals.

Another option involves using other ordinal-specific dissimilarity measures, such as those based on ranks, monotonic transformations, or agreement/disagreement counts. These measures are designed to avoid the pitfalls of assuming equal spacing between categories.

Once an appropriate dissimilarity matrix is computed (e.g., using Gower's distance), clustering algorithms that do not rely on Euclidean geometry or mean centroids can be effectively employed.

Some examples are:

- Hierarchical clustering: Builds a hierarchy of clusters from individual data points, based on their dissimilarities.
- Partitioning Around Medoids (PAM): A robust alternative to K-means that uses actual data points (medoids) as cluster centers, minimizing the sum of dissimilarities rather than squared Euclidean distances.

3.3.2 Model-Based Clustering

Beyond distance-based approaches, model-based clustering offers a probabilistic framework for identifying clusters. Instead of relying only on distance metrics, model-based clustering assumes that the data points within each cluster are generated from a specific probability distribution, and the entire dataset is a mixture of these distributions. The goal is to estimate the parameters of these component distributions and assign observations to the clusters that they are most likely to belong to.

For ordinal data, model-based clustering often employs specialized distributions that are designed for ordered categories. For example, some approaches might use latent class models, where each latent class represents a cluster, and within each cluster, the ordinal responses follow a specific discrete probability distribution (e.g., a multinomial distribution or a specific ordinal-friendly distribution like the Binary Ordinal Search (BOS) model). This allows for a more rigorous statistical inference on the number of clusters and the characteristics of each cluster.

An important advantage of model-based clustering is its ability to:

- Provide a statistical basis for clustering: This allows for formal tests and information criteria (like BIC or AIC) to select the optimal number of clusters.
- Handle uncertainty in cluster membership: Instead of hard assignments, observations are often assigned to clusters probabilistically.

This approach allows researchers to segment individuals into genuinely homogeneous groups, revealing hidden patterns in the data that might not be evident through classical analyses while also providing a framework for statistical inference about the clustering solution.

Final notes

This course has offered an introduction to the analysis of ordinal data, a type of data that is often used in social sciences, psychology, marketing, and many applied domains.

We have covered the following main topics:

- Definition and characteristics of ordinal data, and the importance of treating them appropriately rather than assuming interval or continuous scale properties.
- Survey and scale design.
- Statistical models for ordinal data, such as the Proportional Odds Model and CUB models.
- Multivariate methods for ordinal and mixed data.

Throughout the course, a common thread has been the need to respect the nature of ordinal data, choosing analytical methods that reflect their structure and meaning. Applying models designed for other types of data may lead to invalid interpretations, while appropriate methods can provide valuable insights into individual preferences, attitudes, and behaviors.